AI -CLOUD AGROSIGHT: COLLABORATION PLATFORM FOR PLANT DISEASE IDENTIFICATION, TRACKING AND FORECASTING

ABSTRACT

The rapid proliferation of plant diseases presents a critical threat to global food security, with nearly 35% of field crops in India being lost annually to pathogens and pests. Traditional diagnostic methods rely on manual inspection by agricultural experts, whose limited availability in rural regions often leads to delayed detection and the subsequent indiscriminate use of toxic chemical pesticides. To bridge this gap, we present **AI-Cloud AgroSight**, a pioneering integrated collaborative platform for automated plant disease identification, tracking, and forecasting. By leveraging a mobile-accessible web interface, farmers can capture images of affected crops for instant, real-time diagnosis powered by cloud-based **Convolutional Neural Networks (CNN)**. The system features a continuous learning mechanism that incorporates expert feedback and user data to enhance predictive accuracy, which has already achieved over 95% in experimental trials. Furthermore, the platform utilizes geo-tagged data to render disease density maps and spread forecasts, enabling proactive regional management and promoting ecologically sustainable agricultural practices.

EXISTING SYSTEM

The current framework for managing crop health is predominantly reliant on manual inspection, where the identification of issues depends on the visual observation of external symptoms such as wilting, lesions, or discoloration. This traditional approach creates a heavy expert dependency, as a reliable and accurate diagnosis typically requires a trained plant pathologist to physically visit the farm site to examine the crops. Because of the lack of immediate diagnostic tools, farmers often resort to preventative chemical use, applying excessive amounts of pesticides as a broad measure to protect their yields against perceived threats.

DISADVENTAGES OF EXISTING SYSTEM

The major limitations of the existing system are as follows:

- **Scarcity of Experts:** Pathologists are often inaccessible to farmers in remote geographical areas.
- **Economic Loss:** Delayed identification allows infectious diseases to spread, resulting in total crop yield failure.
- **Environmental Toxicity:** Indiscriminate chemical use leads to soil degradation and human health risks through bioaccumulation.
- **No Early Warning:** The current setup lacks any mechanism for real-time tracking or regional forecasting.

PROPOSED SYSTEM

The proposed system introduces an AI-driven diagnosis model that utilizes Convolutional Neural Networks (CNN) trained on large, self-collected datasets to classify plant diseases with high precision. To ensure global accessibility and power, the platform employs cloud-centric scaling, where centralized cloud services handle compute-heavy tasks and store diagnostic history. Users interact with the system via web-based access; a Python-based web app allows for universal connectivity without the need for high-end hardware or complex installations. Additionally, the system incorporates geospatial tracking, which automatically extracts location data from uploaded images to visualize disease outbreaks on interactive regional maps.

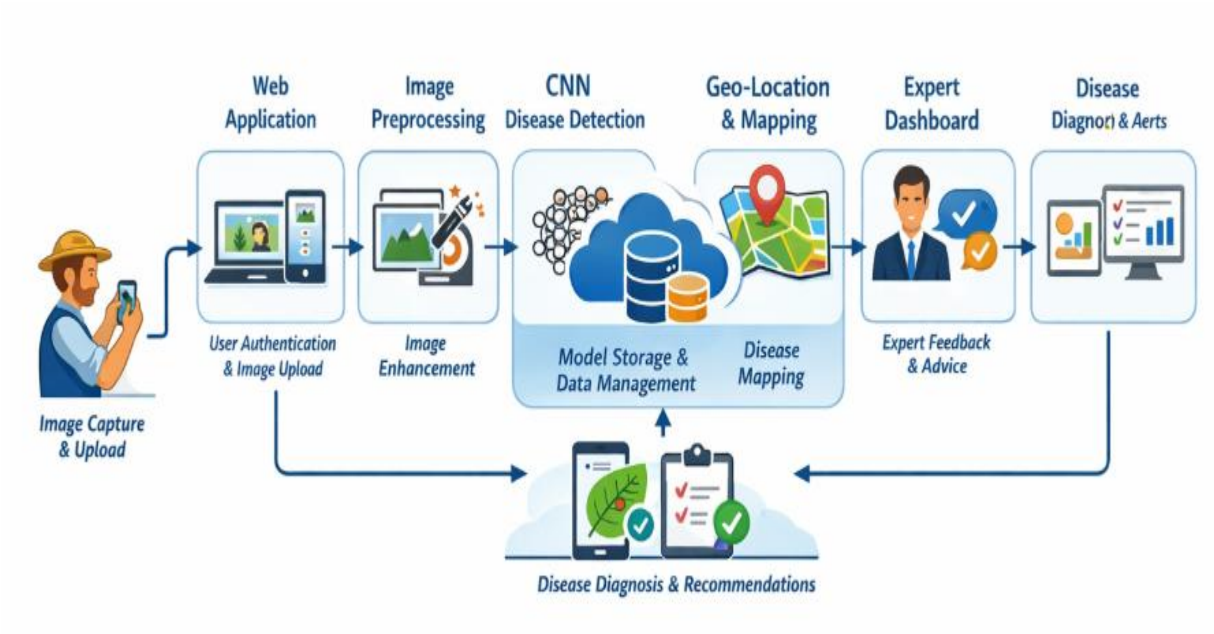
ADVANTAGES OF PROPOSED SYSTEM

The proposed system offers the following advantages:

- **High Accuracy:** Achieves over 95% identification accuracy using deep learning.
- **Scalability:** Accessible from any remote location with internet connectivity.
- **Sustainability:** Reduces chemical dependency by recommending targeted treatments for specific diseases.
- **Expert Collaboration:** Facilitates a bridge between farmers and specialists for validated decision-making.

SYSTEM ARCHITECTURE

The system is built on a modular, three-tier framework designed to ensure both operational efficiency and long-term scalability. It begins with a User Interface Layer, which provides a streamlined web portal for farmers to manage their profiles and upload crop images for analysis. These images are then handled by the Application Processing Layer, where they undergo essential preprocessing—including resizing, normalization, and noise reduction—to prepare the data for the AI Disease Identification Module. This core "intelligence" unit utilizes a trained Convolutional Neural Network (CNN) to accurately classify crop diseases. Supporting these operations is the Cloud Service Layer, which acts as the system's backbone by managing databases, hosting trained models, and performing geospatial analytics. Finally, the system includes an Expert Interface, a specialized dashboard that allows agricultural specialists to validate AI predictions and continuously update the knowledge base for improved accuracy.



SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS:-

- **System** : Pentium IV 2.4 GHz.
- **Hard Disk** : 40 GB.
- **Floppy Drive** : 1.44 Mb.
- **Monitor** : 15 VGA Colour.
- **Mouse** : Logitech.
- **Ram** : 512 Mb.

SOFTWARE REQUIREMENTS:

- **Operating System:** Windows
- **Coding Language**: Python 3.7